

SUPERVISED MACHINE LEARNING REGRESSION STUDY FOR AQUEOUS SOLUBILITY

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Introduction

Solubility is a crucial factor in deciding a drug's bioavailability in pharmaceutical preclinical trials¹. Efficiently predicting solubility during early-stage in silico screenings of hit compounds can significantly reduce the time and costs traditionally associated with synthesizing and testing numerous compounds². Making supervised machine learning an invaluable tool, utilizing labelled data, where compounds are paired with known solubility values³. Aqueous solubility is a continuous variable so supervised regression analysis was the ideal approach. The goal of this study is to model and predict solubility using various machine learning algorithms, comparing their performance to find the most effective model in predicting solubility, then evaluating, and validating the top model. Ensuring the precision of the predictions and their practical applicability in drug development.

Materials and Methods

Materials

The data was downloaded as a version of the dataset created in this 2019 publication². Software used for this analysis included R studio, Python via both Jupiter Notebook and google collab.

Methods

The first step was to assign the features and target (solubility) variables before splitting the data into 3 parts using the train test split function in the sklearn⁴ model selection package. The training dataset had a total of 1337 entries, 75% of the data, while the validation and the test datasets were 25% (446) and 10% (199), respectively. Followed by scaling the data using the StandardScaler function in the sklearn preprocessing package. The training dataset was used for the fitting then both the training and testing datasets are transformed into the new scale.

Model building

Various regression models were compared; a list of the models is in Table 1.

Table 1: Models and ensemble models tested on the dataset:

Models	Ensemble models
Linear regression (LR)	Extra Tree Regression (ET)
Random forests regression (RF)	Stacking CV Regression (includes all models (+ET), using the lasso model as the meta regressor) ⁵
Support vector regression (SVR)	Bagging Regression (for ET)
K-nearest neighbour (KNN)	AdaBoost (for ET)
Linear RANdom SAmple Consensus regression (ransac)	Light gradient boosting
Kernel Ridge Regression (KRR)	eXtreme Gradient Boosting (XGB)

Model comparison using Mean absolute percentage error (MAPE), Root mean squared error (RMSE), Mean squared error (MSE), and Coefficient of determination (R2) (the equations are available in Figure A1⁶). R2 may not be the ideal candidate for evaluating the model's performance, however, it is the easiest to interpret for a broader audience, without a need for an advanced statistical course⁷. Therefore, throughout the graphs the prediction score is presented in R2. The higher the R2, the better the model's performance. Nevertheless, in the table of results the other more appropriate error metrics are mentioned for both the training and the testing datasets.

Visualization of model performance was achieved with Residual plots, Learning curves, Prediction interval graphs, Validation curves, and Feature importance graph using SHAP⁸. These plots may be found throughout the text for LR and ET models, however, for some more models, their graphs are in Figures A4, A5, and A6.

Tuning hyperparameters for ET

Since the main dataset is divided into three parts, the validation dataset's main purpose is to help in tuning the hyperparameters of the model so that the model is tuned not on the training dataset to avoid bias and model overfitting. Tuning was done solely for the best performing model, ET, to optimise its performance. The tuning was carried out within cross-validation, so first the validation dataset was divided into 12 parts (K folds), then the model was trained on a mix of the fold and one-fold was kept assessing the predictivity of the model. Figure A3 illustrates the optimal number of trees for the model, the ideal tree count is 500 due to its smaller standard deviation and high R2 value.

Results and Discussion

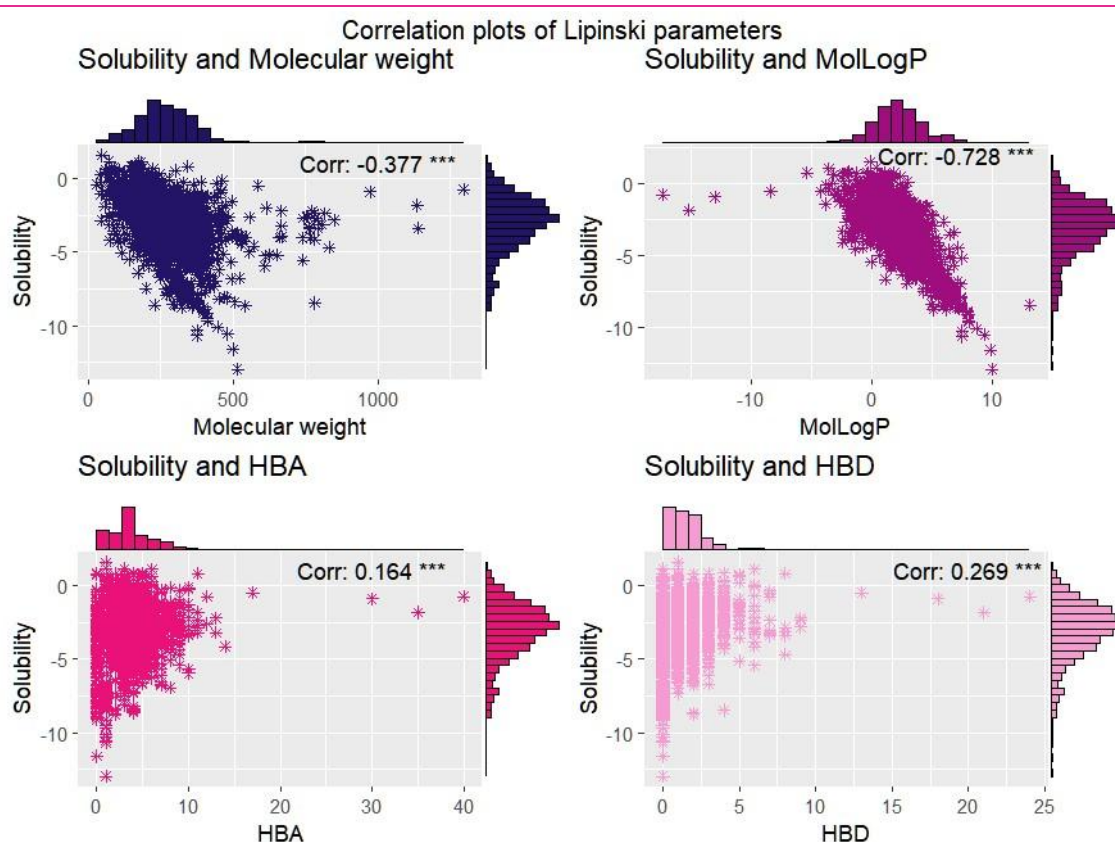


Figure 1: The correlation and distribution of the Lipinski parameters with solubility.

Correlation analyses between solubility and the Lipinski parameters in Figure 1 show negative correlation with MLogP (-0.728), indicating that increased lipophilicity is associated with decreased aqueous solubility. Molecular weight was also inversely related to solubility (-0.377), proving the trend of larger molecules having lower solubility. Moreover, both the number of hydrogen bond acceptors (HBA) and donors (HBD) displayed positive correlations with solubility reflecting the role of hydrogen bonding in enhancing solubility in aqueous solutions. A correlation matrix of solubility and all descriptors is in Figure A2.

Table 2: Model comparisons

Method	Training RMSE	Training MSE	Training R2	Test RMSE	Test MSE	Test R2
Bagging et	0.37701	0.14214	0.96238	0.81326	0.66139	0.85237
ET	0.01699	0.00029	0.99992	0.81778	0.66876	0.85072
ADA	0.16178	0.02617	0.99307	0.81901	0.67077	0.85028
XGB	0.10974	0.01204	0.99681	0.90126	0.81226	0.81869
Random forest	0.89092	0.79373	0.78991	0.90779	0.82409	0.81605
Light GB	0.10022	0.01005	0.99734	0.92751	0.86028	0.80797
SVR	0.95210	0.90649	0.76006	0.94123	0.88591	0.80225
LR	1.12980	1.27645	0.66214	1.02602	1.05272	0.76502
Stack Lasso	0.66729	0.44528	0.88214	1.04487	1.09176	0.75630
KNN	0.02403	0.00058	0.99985	1.19505	1.42813	0.68122
KRR	0.25999	0.06760	0.98211	2.06705	4.27269	0.04628
Linear RANSAC	2.88976	8.35074	-1.21037	2.96253	8.77661	-0.95906

The best performing model was ET for the training dataset with a training MSE score of 0.00029, as seen in table 2. While for the unseen, testing dataset, a combination of ET in a bagging ensemble model gave a lower MSE score. Figure 2 confirms the high R2 value (0.772), implying a strong connection between predicted and actual values. Points closely line up on the identity line, implying precise predictions.

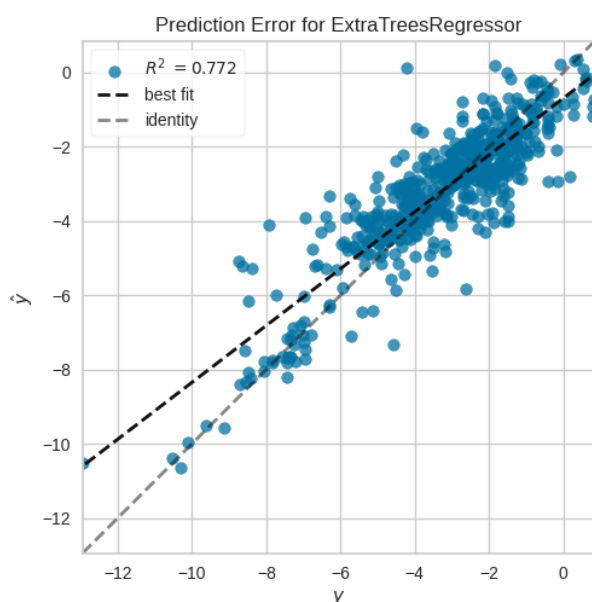


Figure 2: Prediction Error for the Extra Trees Regression model. On the x axis lies actual solubility while the y axis displays predicted solubility.

ET and bagging with ET:

ET is an ensemble learning model, that handles complex, non-linear associations between descriptors and the target variable⁹. It forms an array of decision trees during training and gives the average prediction of the individual trees as an output. This method lowers the model's variance, increasing its generalizability. Benefiting from all the advantages of ET, bagging with ET refines the predictive capability. Bagging, trains multiple samples of the same model on varying subsets of the training data and then mixes their predictions¹⁰. This method reduces overfitting significantly, improves the accuracy, and provides a more consistent solubility prediction by averaging out biases and variances in individual trees.

Feature importance

The significance of the molecular descriptors used in building the ET model can be seen in Figure 3 and compared with different models in Figure A4. MolLogP is a measure of molecular permeability, that is highly dependent on lipophilicity, emerged as the feature with the greatest model impact, showing the highest SHAP values¹¹. An oral drug with a high bioavailability must be both permeable and soluble. However, there is a trade off between the two characteristics, as highly soluble compounds often the case have a high total polar surface area, decreasing their lipophilicity and their ability to permeate the cells with ease. Therefore, MolLogP is seen in Figure 3 to have a higher value of negative impact on the model's ability in predicting solubility.

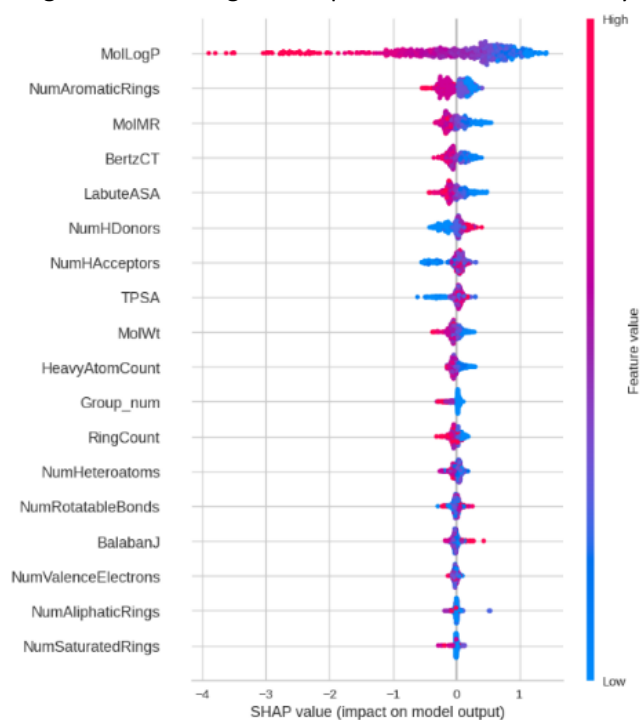


Figure 3: The ranked feature importance and value for the Extra Tree Regression model. The higher the value of the feature the redder it is. This plot reinforces the model's predictive decisions, giving insights into which properties are most influential in predicting solubility.

Validation

A validation curve (Figure 4) was plotted to determine the model's level of complexity and to ensure the balance between the model's generalizability and its ability to learn from the training data. Additionally, the curve illustrates the positive association of the maximum depth hyperparameter and an improved R2 score. Indicating that deeper trees build better ET models.

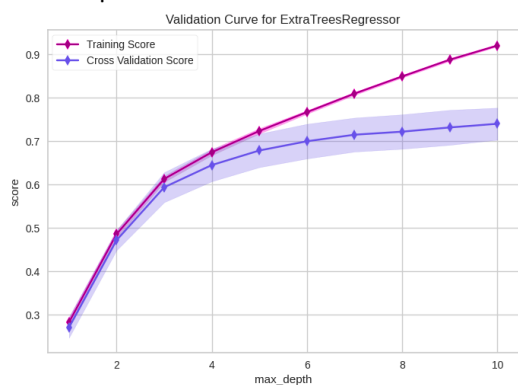


Figure 4: Validation curve for the Extra Trees Regression model

PyCaret:

Validation using PyCaret¹² open-source machine learning package in python with session ID 55. This package analysed a library of models, including models that were analysed in this study and it was able to prove that ET is the best model for this dataset. The model's performance as well as parameters are seen in Figure 5.

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
et	Extra Trees Regressor	0.6933	0.9370	0.9671	0.7398	0.2778	0.8440	0.7850
lightgbm	Light Gradient Boosting Machine	0.7274	0.9380	0.9671	0.7398	0.2778	0.8440	0.8300
rf	Random Forest Regressor	0.7308	0.9583	0.9768	0.7350	0.2785	0.8679	0.9960
xgboost	Extreme Gradient Boosting	0.7312	0.9723	0.9837	0.7313	0.2873	0.8983	0.1980
gbr	Gradient Boosting Regressor	0.7623	1.0014	0.9986	0.7230	0.2845	0.9496	0.3200
ada	AdaBoost Regressor	0.8744	1.2389	1.1115	0.6570	0.3126	1.1344	0.1630
br	Bayesian Ridge	0.9022	1.3540	1.1609	0.6237	0.3278	1.0077	0.0380
lr	Linear Regression	0.9015	1.3546	1.1610	0.6236	0.3280	1.0004	0.0340
ridge	Ridge Regression	0.9015	1.3545	1.1610	0.6236	0.3279	1.0006	0.0310
huber	Huber Regressor	0.9027	1.3905	1.1779	0.6116	0.3343	0.9936	0.0900
lar	Least Angle Regression	0.9212	1.4067	1.1814	0.6084	0.3344	1.0096	0.0580
knn	K Neighbors Regressor	0.9876	1.7321	1.3148	0.5204	0.3544	1.1439	0.0320
en	Elastic Net	1.0450	1.7645	1.3262	0.5122	0.3555	1.2306	0.0390
dt	Decision Tree Regressor	1.0111	1.8836	1.3697	0.4780	0.3927	1.0594	0.0510
lasso	Lasso Regression	1.1371	2.0708	1.4366	0.4283	0.3821	1.3158	0.0340
llar	Lasso Least Angle Regression	1.1371	2.0709	1.4366	0.4283	0.3821	1.3159	0.0390
omp	Orthogonal Matching Pursuit	1.3780	3.2148	1.7906	0.1141	0.4447	1.7644	0.0330
dummy	Dummy Regressor	1.4748	3.6827	1.9160	-0.0121	0.4877	2.0917	0.0170
par	Passive Aggressive Regressor	1.5459	4.3045	2.0045	-0.2068	0.4623	1.5038	0.0340

Parameters	Settings
bootstrap	FALSE
ccp_alpha	0
criterion	squared_error
max_depth	None
max_features	1
max_leaf_nodes	None
max_samples	None
min_impurity_decrease	0
min_samples_leaf	1
min_samples_split	2
min_weight_fraction_leaf	0
n_estimators	100
n_jobs	-1
oob_score	FALSE
random_state	55
verbose	0
warm_start	FALSE

Figure 5: On the left a table comparing various models' performance produced by PyCaret. On the right a table displaying the settings of different parameters to receive a reproducible result.

Quantifying uncertainty

ET model was run for 1000 runs without setting a random state to check for its overall performance. The results shown in Figure 6, proved that the model's performance continued to be on top. As the same analysis was run for the other models, however running this analysis had a significant limitation, the computational power. It was time-consuming and required a huge amount of computer RAM to be performed as a loop. Google collab rejected this analysis and it had to be performed locally. Therefore, the ensemble models were not run 1000 times, that includes the Bagging ET model, as it was computationally difficult. Another strategy that was used throughout the study was not using random states on purpose to check concurrently the effect of randomness on the model's performance. This however, lead some of the graphs to have an inconsistent R2 values, as the analysis was run on different days.



Figure 6: Violin plots with superimposed box plots showing the distribution of the error metrics for the Extra trees regression model after running it for 1000 times.

Conclusion

For this dataset, the Extra trees regression model has proved to be superior to other supervised regression machine modelling algorithms. Even on combination in different ensemble models, ET proved to be the highest performer. The implementation of machine learning algorithms into the stages of drug development significantly accelerates the initial stages through the reduction of time spent on physical testing and the financial burden.

Appendix

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$
$$\text{MSE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} (y_i - \hat{y}_i)^2$$
$$\text{MAPE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \frac{|y_i - \hat{y}_i|}{\max(\epsilon, |y_i|)}$$

Figure A1: Equations for the metrics used in model performance quantification, where, y is the actual value of solubility, and $\hat{y}$ is the predicted value of solubility.

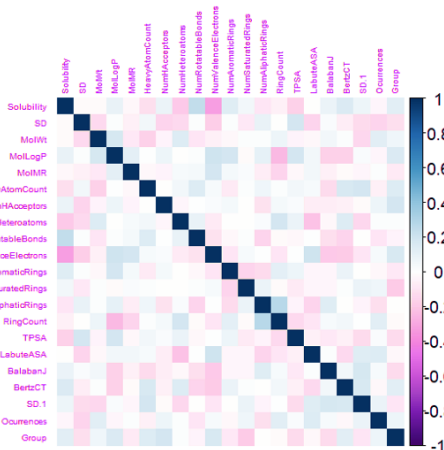


Figure A2: Descriptors correlation plot.

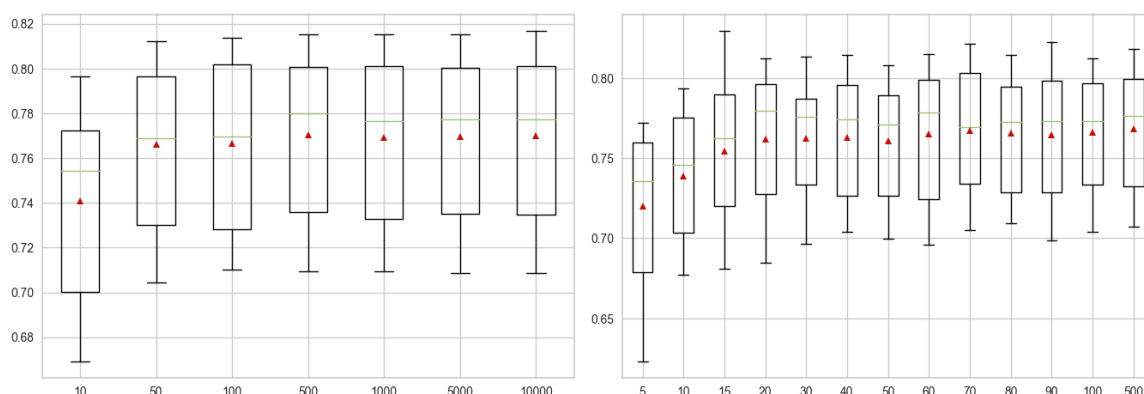


Figure A3: Hyperparameter tuning for the Extra trees regression model, for the number of trees from 10 to 5000, on the left panel, and 5 to 500 on the right panel.

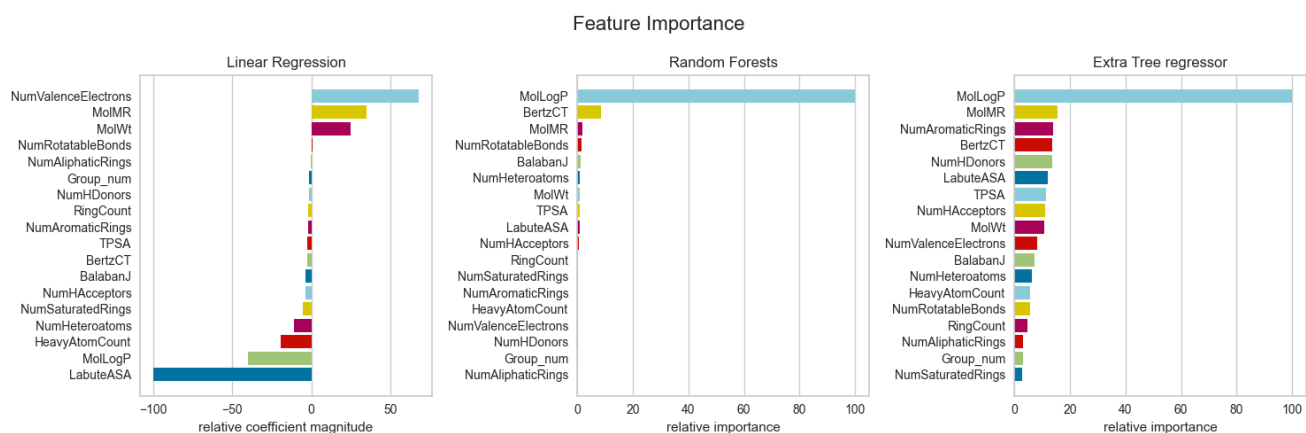


Figure A4: A comparison of feature importance for different models.

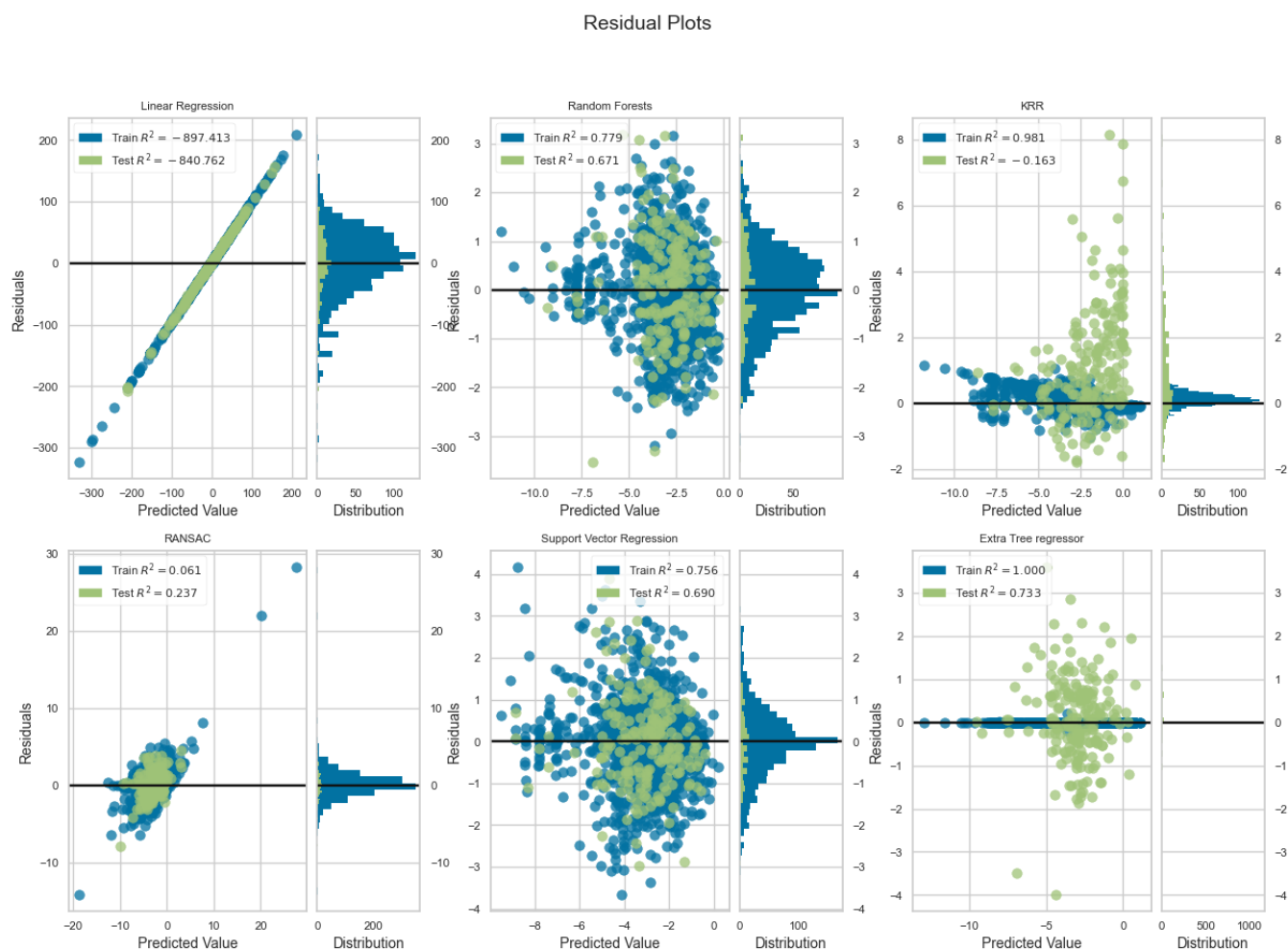


Figure A5: Residual plots for LR, RF, KRR, RANSAC, SVR, and ET

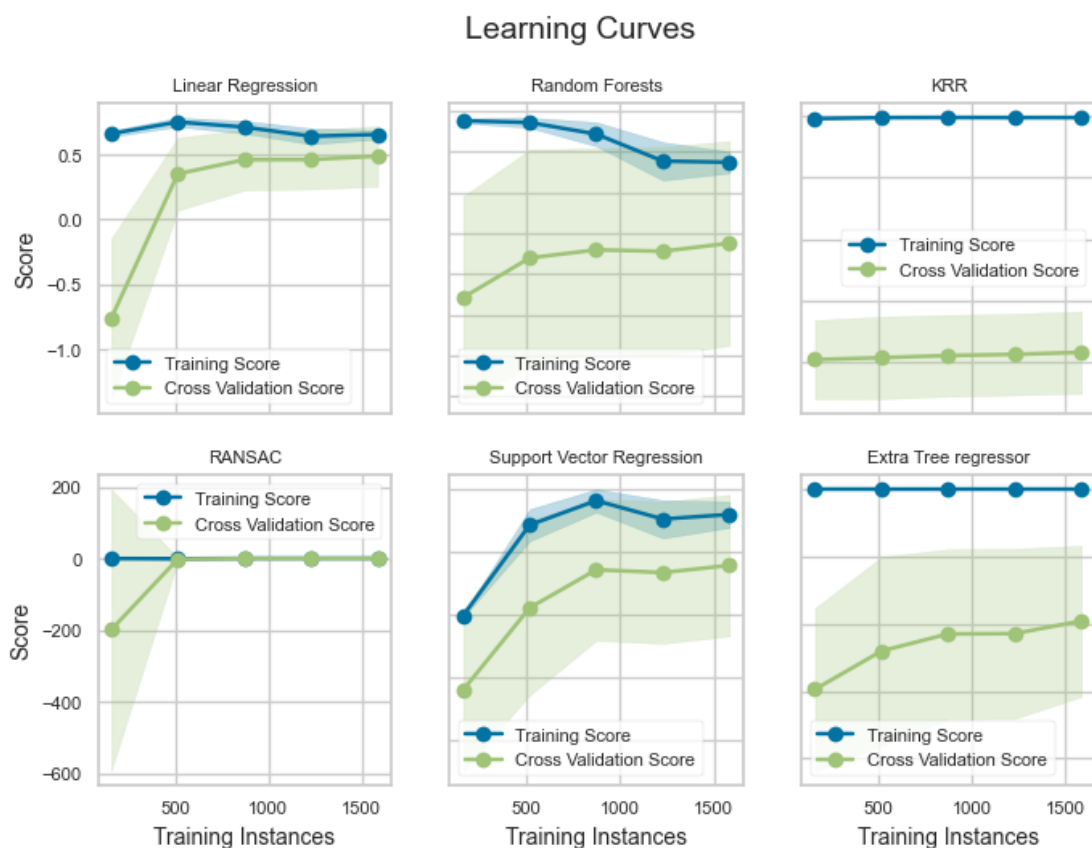


Figure A6: Learning curves for LR, RF, KRR, RANSAC, SVR, and ET

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